Lecture

Tuesday, 21 February 2023 12:02 PM

**Steady State** $\tilde{v}(\rho)$ ; $\nabla^2 v = -\beta\phi^2 G_1(\tilde{v})$ $\quad u = \frac{1+\beta-v}{\beta}$

(Non-isothermal)

$v = \tilde{v} + y$ (perturbation)

↳ We want to see if

reducing dimensionality (don't have to care about $\beta$)

$$\begin{cases} -\nabla^2 y + g(\rho) y = \frac{\partial y}{\partial \tau} \\ BC's: \ \frac{\partial y(0,\tau)}{\partial \rho} = 0 \ ; \ y(1,\tau) = 0 \end{cases} \quad ; \quad g(\rho) = -\beta\phi^2 G_1'(\tilde{v}(\rho)) \leftarrow \text{Linearized problem}$$

Initial boundary value problem

solving eigenvalue problem (using BC & orthogonality)

$$y(\beta,\tau) = \sum_{j=1}^{\infty} a_j z_j(\rho)\, e^{-\mu_j \tau}$$

Perturbation solution

→ $a_j$, $z_j$ are still unknown

$\mu_j$ cannot be negative (or else it will blow up)

If $\mu_j < 0$, $\tilde{v}$ is unstable

**Sturm Comparison Theorem** (Looking for instability)

(We want unstability so $\mu$ should be negative)

Eigenvalue problem: $-\nabla^2 u + g(\rho)u = +\mu u$

BC: $u'(0) = 0$, $u(1) = 0$ } homogeneous BC

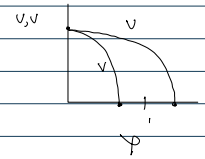
Given an IVP: $V(\rho)$,

$-\nabla^2 V(\rho) + g(\rho)V = 0 \quad 0 < \rho < 1$

$V(0) = 1, \ V'(0) = 0$

This dictates how much it goes up & down

$$\begin{cases} -\nabla^2 v + \underbrace{g(\rho)}_{\text{this dictates how much it goes up \& down}} v = \mu v \\ v(0) = 1, \ v'(0) = 0 \end{cases}$$




$v$ & $V$ have the same no. of zeros

According to Sturm → given, $i^{th}$ zero of $V$ → $i^{th}$ zero of $v$ will be on the right.

∴ Suppose $V$ has a zero in $(0,1)$; then there must be a negative $\mu$ s.t. $v(1) = 0$ (i.e. $v$ has a root at 1) } This is different from a BC since it applies to the solution

↳ This marks the limit of stability for $\tilde{v}$

↳ So then now we need condition for $V$ which allows it to have a zero in $(0,1)$